

Egyptian Russian University

Faculty of Artificial Intelligence

Department of [Artificial Intelligence / Data Science / Cybersecurity]

Graduation Project Report Template

نموذج تقرير مشروع التخرج

Project Title in English

عنوان المشروع باللغة العربية

Submitted to the Department of [Department Name], Faculty of Artificial Intelligence, Egyptian Russian University,
in partial fulfillment of the requirements for the Bachelor's Degree in [Program Name].

Student Name	Student ID	Program/Specialization
[Student 1]	[ID]	[AI / DS / Cybersecurity]
[Student 2]	[ID]	[AI / DS / Cybersecurity]
[Student 3]	[ID]	[AI / DS / Cybersecurity]
[Student 4]	[ID]	[AI / DS / Cybersecurity]

Supervisor(s)	Academic Title and Specialization
[Supervisor Name]	[Title / Department]
[Co-supervisor Name, if any]	[Title / Department]

Academic Year: [20XX/20XX] | Term: [First / Second]

List of Contents

Document Control and Formatting Instructions	4
FINAL SUBMISSION CHECKLIST	6
Declaration.....	6
Supervisor Approval.....	7
Acknowledgements.....	7
Abstract.....	8
List of Abbreviations	10
List of Figures.....	10
List of Tables	10
CHAPTER 1: INTRODUCTION.....	11
1.1 Background.....	11
1.2 Problem Statement.....	11
1.3 Project Motivation	11
1.4 Project Aim and Objectives	11
1.5 Scope and Limitations	11
1.6 Project Contributions	11
1.7 Report Organization.....	11
CHAPTER 2: BACKGROUND AND RELATED WORK.....	12
2.1 Theoretical Background.....	12
2.2 Review Methodology.....	12
2.3 Related Studies and Existing Systems	12
2.4 Comparative Analysis.....	12
2.5 Research/Technical Gap	12
CHAPTER 3: PROJECT ANALYSIS AND REQUIREMENTS.....	13
3.1 Stakeholders and Target Users	13
3.2 Feasibility Study	13
3.3 Functional Requirements	13
3.4 Non-Functional Requirements.....	13
3.5 Data Requirements.....	13
3.6 Risk Analysis	13
3.7 System Analysis Diagrams	13

CHAPTER 4: METHODOLOGY AND SYSTEM DESIGN	14
4.1 Overall Methodology	14
4.2 System Architecture.....	14
4.3 Data Pipeline and Preprocessing	14
4.4 Model/Algorithm/Tool Design	14
4.5 User Interface and User Experience Design	14
4.6 Database and Storage Design	14
4.7 Security, Privacy, and Ethical Design	14
4.8 Specialization-Specific Design Requirements.....	14
CHAPTER 5: IMPLEMENTATION	15
5.1 Implementation Overview	15
5.2 Tools, Technologies, and Frameworks	15
5.3 Development Environment.....	15
5.4 Module Implementation.....	15
5.5 Integration.....	15
5.6 Deployment or Prototype Setup.....	15
5.7 Challenges and Solutions.....	15
CHAPTER 6: TESTING, EVALUATION, AND RESULTS	16
6.1 Testing Objectives	16
6.2 Testing Environment	16
6.3 Test Cases	16
6.4 Evaluation Metrics.....	16
6.5 Results Presentation.....	16
6.6 Discussion of Results.....	16
CHAPTER 7: PROJECT MANAGEMENT, ETHICS, AND SUSTAINABILITY	17
7.1 Work Breakdown Structure	17
7.2 Project Timeline.....	17
7.3 Team Roles and Contributions	17
7.4 Ethical and Legal Considerations	17
7.5 Sustainability and Societal Impact.....	17
CHAPTER 8: CONCLUSIONS AND FUTURE WORK	18
8.1 Conclusion	18
8.2 Limitations	18

8.3 Future Work.....	18
REFERENCES	18
APPENDICES	18
Appendix A: User Manual.....	18
Appendix B: Dataset Description or Data Dictionary	18
Appendix C: Additional Results.....	18
Appendix D: Essential Code Snippets.....	19
Appendix E: Cybersecurity Permission Statement, if applicable.....	19

Document Control and Formatting Instructions

Item	Required Format
Font	Times New Roman. Main text: 12 pt; chapter titles: 16 pt; section titles: 14 pt; subsection titles: 13 pt.

Item	Required Format
Page layout	A4 paper; margins approximately 2.0-2.2 cm; justified text; 1.15 or 1.5 line spacing according to department requirements.
Citation style	IEEE style is recommended for computing disciplines. Use one citation style consistently throughout the report.
Figures and tables	Every figure and table must be numbered, titled, cited in the text, and placed close to its first mention.
Code and datasets	Include essential code snippets only. Lengthy code, logs, datasets, or manuals should be placed in appendices or submitted separately.
Ethics	Projects using human data, sensitive data, medical data, or security testing must include ethical approval/consent and data protection measures.

Important: This template is designed for graduation projects in Artificial Intelligence, Data Science, and Cybersecurity. Students should keep the main structure while selecting the specialization-specific sections that match their project.

FINAL SUBMISSION CHECKLIST

No.	Checklist Item	Done
1	Project title, team members, supervisor, program, and academic year are complete.	☐
2	Abstract is concise and includes objective, methodology, results, and contribution.	☐
3	All chapters are complete and aligned with the project type.	☐
4	All figures and tables are numbered, captioned, and cited in the text.	☐
5	Requirements, methodology, implementation, testing, and results are consistent with each other.	☐
6	Specialization-specific requirements for AI, Data Science, or Cybersecurity are addressed.	☐
7	Ethics, privacy, data protection, and security permissions are addressed where applicable.	☐
8	References are complete, recent, relevant, and formatted consistently in IEEE style.	☐
9	Language, grammar, spelling, and formatting have been reviewed.	☐
10	Final Word/PDF files and supplementary materials are submitted according to faculty instructions.	☐

Declaration

We, the undersigned, declare that the work presented in this graduation project report titled “[Project Title]” is our original work carried out under the supervision of [Supervisor Name]. The report has not been submitted elsewhere for any academic degree or qualification. All sources, datasets, tools, libraries, and external contributions have been properly acknowledged and cited. We confirm that the

project complies with the academic integrity, research ethics, data protection, and professional conduct regulations of the Egyptian Russian University.

Student Name	Student ID	Department	Signature	Date

Supervisor Approval

This is to certify that the graduation project report has been reviewed and approved for submission.

Supervisor Name	Signature	Date	Comments

Acknowledgements

Write a concise acknowledgement paragraph thanking Allah, the supervisor(s), faculty members, colleagues, family, and any supporting institution. Avoid informal language and keep the section professional and brief.

Abstract

The abstract should be 200-300 words and should not include citations, figures, or tables. It should summarize the project background, problem, objective, methodology, key results, and main contribution.

Abstract Component	Guiding Questions
Background and Problem	What real problem does the project address? Why is it important?
Objective	What is the main aim of the project? What will the system/model/tool achieve?
Methodology	What approach, algorithm, dataset, tools, architecture, or security method was used?
Results	What are the main measurable outcomes? Use accuracy, F1-score, RMSE, response time, detection rate, dashboard insights, or security metrics as applicable.

Abstract Component	Guiding Questions
Contribution	What is new, useful, or practically significant about the project?

List of Abbreviations

Abbreviation	Meaning
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
NLP	Natural Language Processing
DS	Data Science
EDA	Exploratory Data Analysis
API	Application Programming Interface
CIA	Confidentiality, Integrity, and Availability
IDS/IPS	Intrusion Detection/Prevention System

List of Figures

List of Tables

CHAPTER 1: INTRODUCTION

1.1 Background

Introduce the domain, real-world context, target users, and why the topic is important for Artificial Intelligence, Data Science, or Cybersecurity.

1.2 Problem Statement

Define the problem precisely. Avoid broad statements. Explain who is affected, what limitation exists, and why current solutions are insufficient.

1.3 Project Motivation

Explain the academic, technical, social, business, or security motivation behind the project.

1.4 Project Aim and Objectives

State one overall aim and 4-6 measurable objectives. Use action verbs such as design, develop, train, evaluate, deploy, compare, secure, visualize, or analyze.

1.5 Scope and Limitations

Define what is included and excluded. Mention assumptions, dataset boundaries, hardware/software constraints, privacy restrictions, and deployment limits.

1.6 Project Contributions

Summarize the main contributions, such as a model, dataset pipeline, dashboard, application, framework, security tool, comparative analysis, or prototype.

1.7 Report Organization

Briefly describe the contents of each chapter.

Objective No.	Objective Statement	Success Indicator / Evidence
O1		
O2		
O3		
O4		

CHAPTER 2: BACKGROUND AND RELATED WORK

2.1 Theoretical Background

Explain the key concepts, algorithms, protocols, datasets, standards, or technologies required to understand the project.

2.2 Review Methodology

Explain how papers, systems, tools, or datasets were selected. Mention databases such as IEEE Xplore, ACM, Springer, ScienceDirect, arXiv, Kaggle, GitHub, NIST, OWASP, or vendor documentation where relevant.

2.3 Related Studies and Existing Systems

Organize the literature thematically by approach, method, domain, or timeline. Summarize each work briefly and objectively.

2.4 Comparative Analysis

Compare at least 6-10 relevant works using clear criteria.

2.5 Research/Technical Gap

Conclude with what is missing and how the proposed project addresses the gap.

Ref.	Author/Year or System	Problem Addressed	Method/Technology	Dataset/Environment	Strengths	Limitations	Relation to Our Project
[1]							
[2]							
[3]							

CHAPTER 3: PROJECT ANALYSIS AND REQUIREMENTS

3.1 Stakeholders and Target Users

Identify users, administrators, decision-makers, data owners, security analysts, or external beneficiaries.

3.2 Feasibility Study

Discuss technical, operational, economic, legal/ethical, data, and schedule feasibility.

3.3 Functional Requirements

List what the system must do. Each requirement should be testable.

3.4 Non-Functional Requirements

Define performance, usability, reliability, scalability, maintainability, privacy, security, accessibility, and interoperability requirements.

3.5 Data Requirements

Specify data sources, variables/features, collection method, volume, quality, labeling, storage, privacy, and access permissions.

3.6 Risk Analysis

Identify risks related to data quality, bias, cyber misuse, implementation complexity, budget, time, and ethical issues.

3.7 System Analysis Diagrams

Include diagrams as applicable: use case diagram, activity diagram, data flow diagram, ERD, class diagram, sequence diagram, data pipeline diagram, or threat model diagram.

Req. ID	Requirement	Type	Priority	Verification Method
FR-01		Functional	High/Medium/Low	Test case / demo
NFR-01		Non-functional	High/Medium/Low	Metric / inspection

Risk	Probability	Impact	Mitigation	Responsible
Data quality issues	Medium	High	Apply cleaning, validation, and documentation.	Team
Privacy or sensitive data exposure	Low/Medium	High	Anonymization, access control, and ethical approval.	Team/Supervisor

CHAPTER 4: METHODOLOGY AND SYSTEM DESIGN

4.1 Overall Methodology

Describe the development/research methodology, such as CRISP-DM, Agile, Design Science, experimental ML pipeline, or secure software development lifecycle.

4.2 System Architecture

Present a high-level architecture diagram showing users, interfaces, backend, database, model/service layer, analytics layer, and deployment environment.

4.3 Data Pipeline and Preprocessing

Explain data collection, cleaning, transformation, labeling, feature engineering, splitting, and validation.

4.4 Model/Algorithm/Tool Design

Explain the selected model, algorithm, analytical method, or security approach and justify the choice.

4.5 User Interface and User Experience Design

Include wireframes/mockups and justify design principles such as simplicity, accessibility, responsiveness, and clarity.

4.6 Database and Storage Design

Present ERD/schema or storage design where applicable. Explain tables, relationships, indexes, access permissions, and retention rules.

4.7 Security, Privacy, and Ethical Design

Explain authentication, authorization, data protection, anonymization, consent, audit logs, responsible AI, and secure coding practices as applicable.

4.8 Specialization-Specific Design Requirements

Specialization	Required Design Elements
Artificial Intelligence	Model architecture, training pipeline, hyperparameters, baseline model, explainability method, model deployment approach, bias/fairness considerations.
Data Science	Data sources, ETL pipeline, EDA, feature engineering, statistical/ML methods, dashboard/storytelling design, insight validation, decision-support mechanism.
Cybersecurity	Threat model, assets, adversary assumptions, attack surface, security controls, secure architecture, test lab setup, vulnerability assessment approach, legal/ethical boundaries.

CHAPTER 5: IMPLEMENTATION

5.1 Implementation Overview

Summarize the implementation phases and the final deliverable.

5.2 Tools, Technologies, and Frameworks

List programming languages, libraries, frameworks, databases, cloud services, security tools, IDEs, and version control systems.

5.3 Development Environment

Specify hardware, operating system, development tools, package versions, and configuration requirements.

5.4 Module Implementation

Describe each module/component and how it was implemented.

5.5 Integration

Explain how frontend, backend, database, model, APIs, dashboards, or security tools were integrated.

5.6 Deployment or Prototype Setup

Explain local, cloud, mobile, web, containerized, or lab deployment steps.

5.7 Challenges and Solutions

Explain important technical challenges and how they were solved.

Component/Module	Purpose	Technology Used	Inputs	Outputs	Status

Tool/Library	Version	Purpose	Justification
Python		Model development/data analysis	Widely used and well-supported

CHAPTER 6: TESTING, EVALUATION, AND RESULTS

6.1 Testing Objectives

State what was tested: functionality, accuracy, performance, usability, security, scalability, robustness, or interpretability.

6.2 Testing Environment

Describe hardware, software, test data, tools, browsers/devices, test network, or cyber range/lab environment.

6.3 Test Cases

Provide clear test cases with expected and actual results.

6.4 Evaluation Metrics

Define the metrics used and why they are appropriate.

6.5 Results Presentation

Present results using tables, graphs, confusion matrices, screenshots, dashboards, logs, or vulnerability reports.

6.6 Discussion of Results

Interpret the results, compare them with expectations or related work, and explain limitations.

Test ID	Test Scenario	Input/Test Data	Expected Result	Actual Result	Pass/Fail
TC-01					
TC-02					

Specialization	Recommended Metrics
Artificial Intelligence	Accuracy, precision, recall, F1-score, AUC, confusion matrix, loss curves, inference time, explainability outputs.
Data Science	Data quality indicators, descriptive statistics, model metrics such as RMSE/MAE/R ² or classification metrics, dashboard usability, insight validation.
Cybersecurity	Detection rate, false positive rate, vulnerability severity, CVSS score, response time, exploit success/failure, log coverage, compliance with secure practices.

CHAPTER 7: PROJECT MANAGEMENT, ETHICS, AND SUSTAINABILITY

7.1 Work Breakdown Structure

Summarize major tasks and responsibilities among team members.

7.2 Project Timeline

Include a Gantt chart or table covering analysis, design, implementation, testing, documentation, and final presentation.

7.3 Team Roles and Contributions

Clarify each member's technical contribution.

7.4 Ethical and Legal Considerations

Discuss plagiarism, citation, consent, privacy, sensitive data, cybersecurity testing permissions, responsible AI, and misuse prevention.

7.5 Sustainability and Societal Impact

Explain how the project supports practical value, social benefit, digital transformation, environmental sustainability, or industry needs.

Team Member	Main Responsibilities	Deliverables	Evidence of Contribution

CHAPTER 8: CONCLUSIONS AND FUTURE WORK

8.1 Conclusion

Restate the objective, summarize the methodology, highlight the main outcomes, and explain the value of the project.

8.2 Limitations

Mention constraints honestly, such as limited dataset size, deployment restrictions, computational limitations, unavailable real users, or limited testing scope.

8.3 Future Work

Suggest realistic extensions such as larger datasets, real-world deployment, improved model, more robust security testing, mobile version, cloud deployment, or integration with institutional systems.

REFERENCES

Use IEEE citation style consistently. Every in-text citation must appear in the reference list, and every reference must be cited in the report. Suggested tools: Zotero, Mendeley, EndNote, or Word References Manager.

IEEE Reference Examples
[1] J. Smith, Introduction to Machine Learning. Springer, 2021.
[2] C. Brown and D. Green, "Advances in neural networks," AI Journal, vol. 45, no. 3, pp. 123-145, 2020.
[3] OWASP Foundation, "OWASP Top 10," Accessed: [date]. [Online]. Available: [URL]

APPENDICES

Appendix A: User Manual

Include installation steps, screenshots, and user instructions.

Appendix B: Dataset Description or Data Dictionary

Include dataset source, feature names, types, descriptions, preprocessing, and permissions.

Appendix C: Additional Results

Include extra graphs, screenshots, confusion matrices, logs, or dashboard outputs.

Appendix D: Essential Code Snippets

Include only important code snippets. Long code should be hosted in an approved repository or submitted separately according to faculty rules.

Appendix E: Cybersecurity Permission Statement, if applicable

For cybersecurity projects, include written approval for any penetration testing, vulnerability scanning, or security experiment. Never test systems without authorization.